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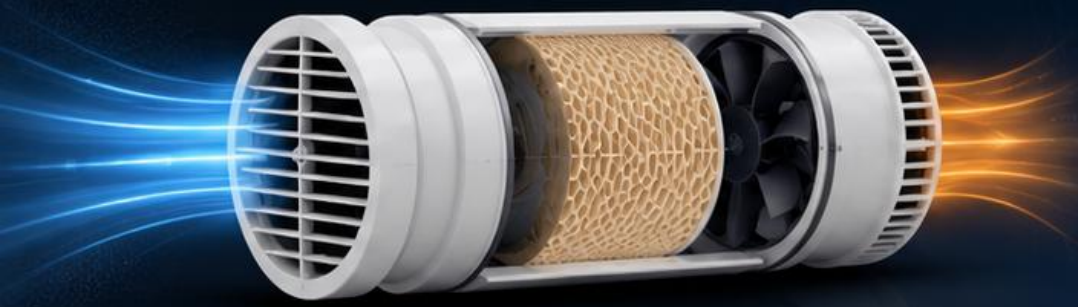
ENGINEERING PORTFOLIO / 2025 – 26

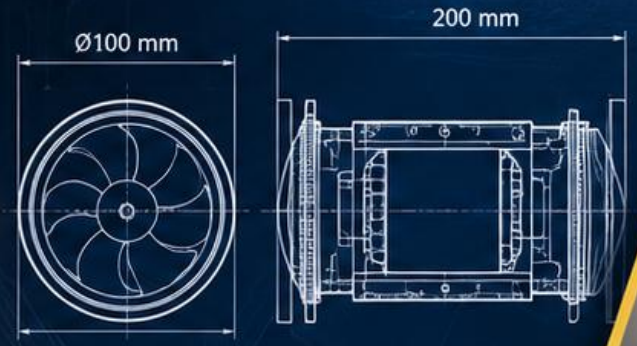
AEROVENT

*A low-cost **heat-recovery** ventilation concept*

Engineering for People Design Challenge | Ladywood, Birmingham

Building a more sustainable and comfortable community







BOKHTEAR UDDIN BADHON
BEng (Hons) Mechanical Engineering
London South Bank University

Team Concept Project
Engineering for People 2025–26

Seeking a **2027 Industrial Placement**

INNOVATIVE ENGINEERING. A SUSTAINABLE TOMORROW.

01 / PROJECT SNAPSHOT

Engineering for people, not just performance

A first-year team challenge investigating energy affordability and housing comfort in Ladywood, Birmingham.

THE CHALLENGE

High heating costs, inefficient housing and the need for fresh air.

THE TEAM CONCEPT

Aerovent: a compact, low-cost heat-recovery ventilation proposal.

MY INITIATIVE

Environmental-context research and an individual solar-sharing proposal.

TEAM OUTCOME / Four alternatives considered; Aerovent selected as the team design direction. Concept only — no prototype validation claimed.

02 / CONTEXT RESEARCH

Understanding Ladywood

The design began with community needs, affordability and environmental constraints.

26.6%

households in fuel poverty

25,500

homes below EPC target C

34.6%

children in poverty

8 m^2

green space per person

Figures reproduced from the submitted team report and its cited sources; they are contextual estimates, not measurements made by our team.

03 / PROBLEM DEFINITION

Fresh air without avoidable heat loss

How might households maintain ventilation while reducing energy lost through existing wall vents?

VENTILATION

Fresh-air movement is important for managing damp and condensation.

THERMAL LOSS

Warm air leaving a home can increase heating demand.

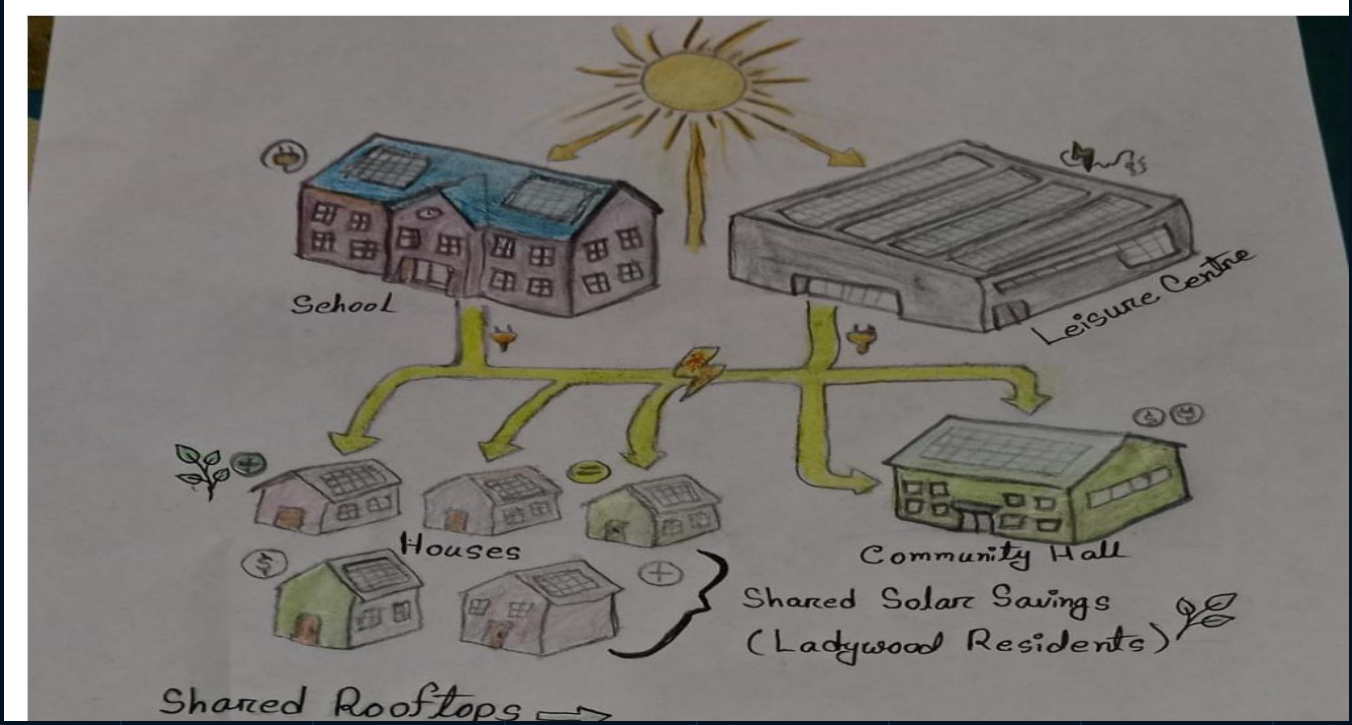
RETROFIT REALITY

Low upfront cost and minimal disruption matter for residents.

04 / MY INDIVIDUAL CONCEPT

Neighbourhood Solar Sharing System

My own Idea 1: community rooftop solar intended to support lower energy costs and shared local benefit.



MY ORIGINAL DRAWING

Solar panels on schools, leisure centres and community buildings; energy and savings shared with nearby households.

An early-stage proposal, not an installed scheme.

05 / DESIGN SELECTION

Four concepts. One team direction.

The team considered solar sharing, canal heat pumps, retrofit glazing and heat-recovery ventilation.

01 / BOKHTEAR

Neighbourhood solar sharing

02 / AMRIT

Canal-powered heat pump

03 / LUIS

Retrofit window glazing

04 / AMOS

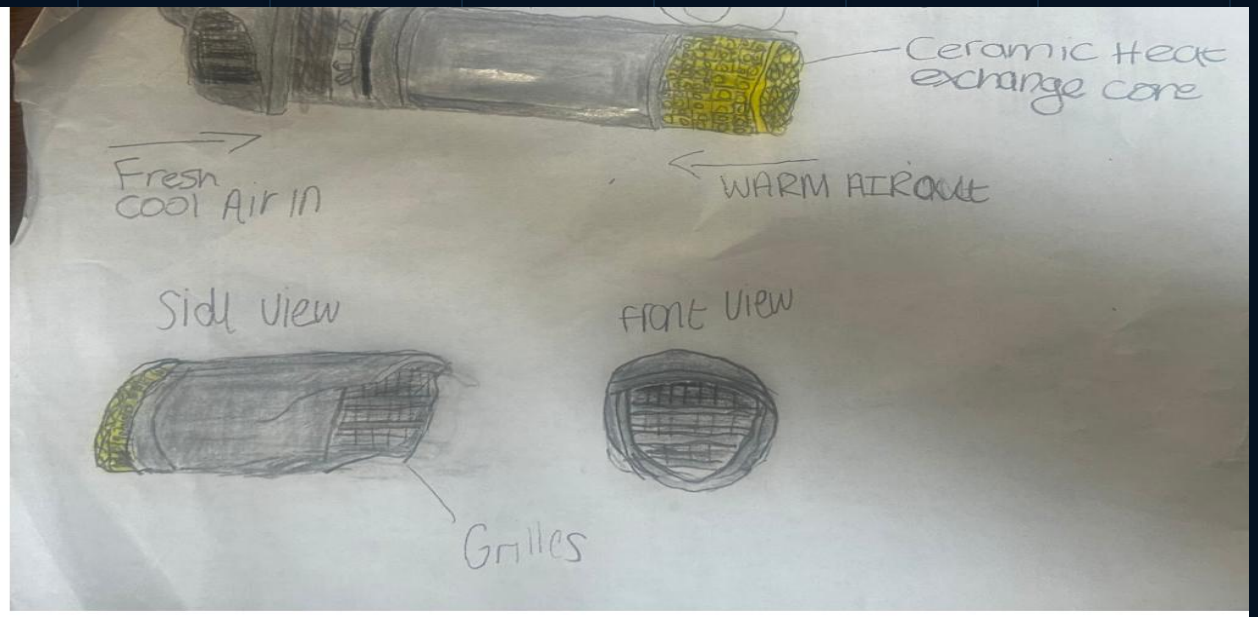
Aerovent heat-recovery ventilator

Selection considered cost, technical feasibility, scalability, use of existing infrastructure, social inclusion and environmental impact.

06 / SELECTED TEAM CONCEPT

AEROVENT / The design proposal

A compact heat-recovery ventilator intended for installation within an existing wall vent.



Idea 4 (Amos) - Ladywood residents tend to have a low household income, so when winter arrives, fuel bills are costly or even worse. Some households aren't able to pay and are then forced to live in colder housing. The Ladywood Aerovent. An inexpensive heat recovery ventilator is designed. The Aerovent is installed within an existing wall vent and utilises a

CORE COMPONENTS

PVC cylindrical housing

Ceramic thermal-storage core

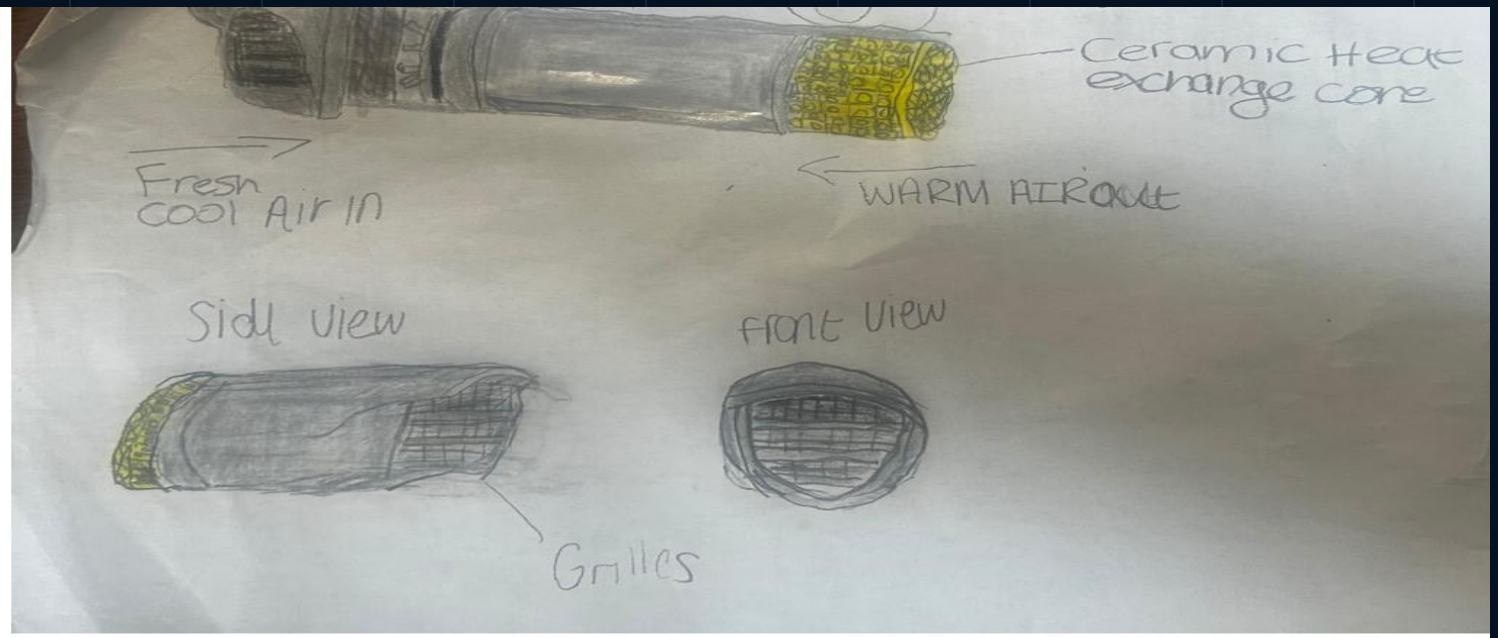
Reversible low-voltage fan

Protective grille and serviceable filter

07 / RAW ENGINEERING SKETCH

From hand drawing to design communication

The team’s original sketch shows airflow direction, ceramic heat-exchange core and grille arrangement.



Idea 4 (Amos) - Ladywood residents tend to have a low household income, so when winter arrives, fuel bills are costly or even worse. Some households aren't able to pay and are then forced to live in colder housing. The Ladywood Aerovent. An inexpensive heat recovery ventilator is designed. The Aerovent is installed within an existing wall vent and utilises a

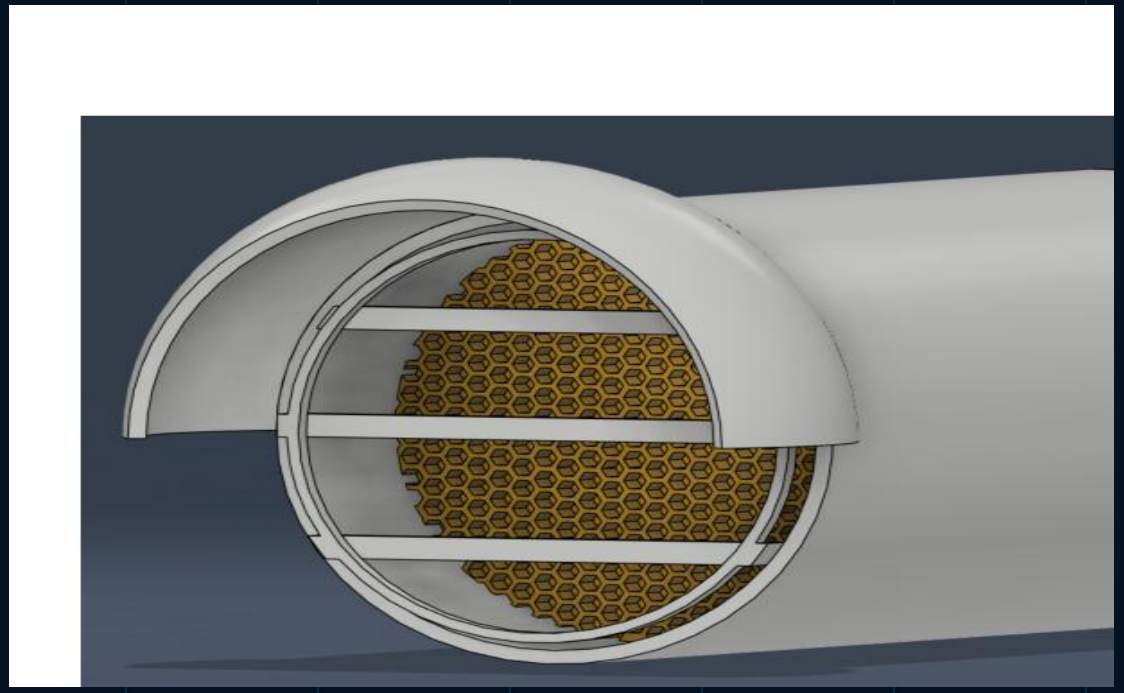
WHY THIS MATTERS

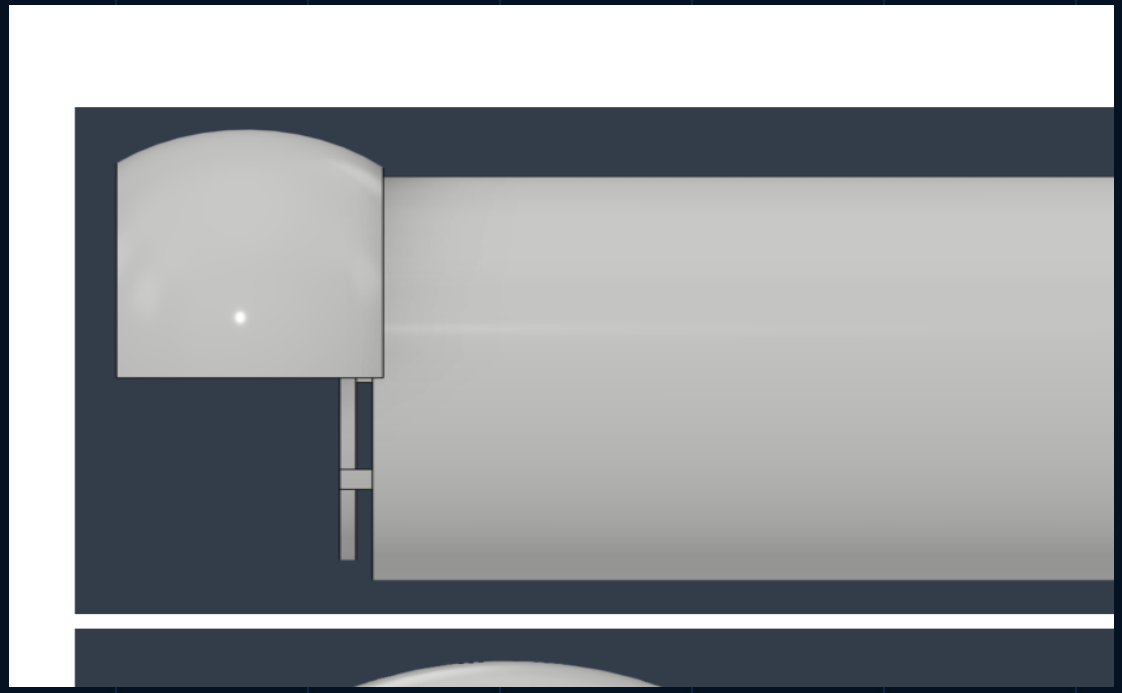
Raw sketches make assumptions visible and allow a concept to be discussed before detailed CAD and testing.

08 / CAD CONCEPT VIEWS

Visualising form and assembly

CAD imagery from the submitted team report helps communicate the cylindrical housing and internal core.





Concept visuals only — not manufacturing drawings, tested prototypes or evidence of individual CAD authorship.

09 / OPERATING PRINCIPLE

Thermal storage through alternating airflow

The ceramic core is intended to capture heat from outgoing air and return some of it to incoming fresh air.

01 / EXHAUST

Warm indoor air travels out

02 / STORAGE

Ceramic core absorbs heat

03 / INTAKE

Fresh outdoor air enters

04 / RELEASE

Core warms incoming air

Heat-recovery efficiency, pressure drop, acoustic performance and airflow rate still require testing.

10 / ENGINEERING CONSIDERATIONS

Materials, maintenance and retrofit

A useful design must be practical to install, affordable to maintain and safe to operate.

PVC HOUSING

Compact cylindrical structure

CERAMIC CORE

Thermal storage and release

GRILLES / FILTERS

Debris protection and cleaning

Proposed installation: existing wall vents where technically suitable; verify sizing, ventilation requirements, noise and access before deployment.

11 / COMMUNITY ENGAGEMENT

Design with residents, not only for them

The report proposes an iterative engagement process; these activities were planned, not completed trials.

01 / LISTEN

Surveys and conversations

02 / CO-DESIGN

Feedback on concept and cost

03 / TRIAL

Small controlled pilot

04 / IMPROVE

Measure and refine

12 / NEXT TECHNICAL STEPS

What would make Aerovent credible?

The strongest next step is evidence-led prototyping rather than claiming unverified performance.

TEST & VALIDATE

- Measure airflow and pressure drop
- Assess noise and heat-recovery performance
- Check safety and wall-vent compatibility

COST & IMPLEMENT

- Estimate component and installation costs
- Assess maintenance and access
- Collect feedback from potential users

13 / FIRST-YEAR REFLECTION

What this challenge developed in me

A practical introduction to responsible engineering: researching a real problem, proposing options and evaluating trade-offs.

MY CONTRIBUTION

Researching Ladywood’s environmental context

Proposing the solar-sharing concept

Contributing to the group design process

SKILLS DEMONSTRATED

Problem framing and research

Concept sketching and communication

Team decision-making

Sustainability and user-centred thinking

14 / INDUSTRIAL PLACEMENT 2027

Looking ahead to engineering practice

Seeking a 2027 industrial placement in London where I can learn from engineers and contribute to meaningful projects.



BOKHTEAR UDDIN BADHON

BEng (Hons) Mechanical Engineering

London South Bank University

Interested in mechanical design, thermal systems, sustainable technology and engineering problem-solving.

Open to placement conversations for 2027.